



6.3.4 Matrix-vector multiplication ¶

6.3.4 Backward error of matrix-vector multiplication

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Assume $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, and $y \in \mathbb{R}^m$. Partition

$$A = \begin{pmatrix} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix} \quad \text{and} \quad y = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{m-1} \end{pmatrix}.$$

Then computing $y := Ax$ can be orchestrated as

$$\begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{m-1} \end{pmatrix} := \begin{pmatrix} \tilde{a}_0^T x \\ \tilde{a}_1^T x \\ \vdots \\ \tilde{a}_{m-1}^T x \end{pmatrix}. \quad (6.3.4)$$

From [R-1B 6.3.3.2](#) regarding the dot product we know that

$$\begin{aligned} \tilde{y} &= \begin{pmatrix} \tilde{\psi}_0 \\ \tilde{\psi}_1 \\ \vdots \\ \tilde{\psi}_{m-1} \end{pmatrix} = \begin{pmatrix} (\tilde{a}_0 + \tilde{\alpha}_0)^T x \\ (\tilde{a}_1 + \tilde{\alpha}_1)^T x \\ \vdots \\ (\tilde{a}_{m-1} + \tilde{\alpha}_{m-1})^T x \end{pmatrix} \\ &= \begin{pmatrix} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix} + \begin{pmatrix} \tilde{\alpha}_0^T \\ \tilde{\alpha}_1^T \\ \vdots \\ \tilde{\alpha}_{m-1}^T \end{pmatrix} x = (A + \Delta A)x, \end{aligned}$$

where $|\tilde{\alpha}_i| \leq \gamma_n |\tilde{a}_i|$, $i = 0, \dots, m-1$, and hence $|\Delta A| \leq \gamma_n |A|$.

Also, from [Corollary 6.3.3.2](#) R-1F regarding the dot product we know that

$$\tilde{y} = \begin{pmatrix} \tilde{\psi}_0 \\ \tilde{\psi}_1 \\ \vdots \\ \tilde{\psi}_{m-1} \end{pmatrix} = \begin{pmatrix} \tilde{a}_0^T x + \tilde{\delta}\psi_0 \\ \tilde{a}_1^T x + \tilde{\delta}\psi_1 \\ \vdots \\ \tilde{a}_{m-1}^T x + \tilde{\delta}\psi_{m-1} \end{pmatrix} = \begin{pmatrix} \tilde{a}_0^T \\ \tilde{a}_1^T \\ \vdots \\ \tilde{a}_{m-1}^T \end{pmatrix} x + \begin{pmatrix} \tilde{\delta}\psi_0 \\ \tilde{\delta}\psi_1 \\ \vdots \\ \tilde{\delta}\psi_{m-1} \end{pmatrix} = Ax + \tilde{\delta}y.$$

where $|\tilde{\delta}\psi_i| \leq \gamma_n |\tilde{a}_i|^T |x|$ and hence $|\tilde{\delta}y| \leq \gamma_n |A| |x|$.

The above observations can be summarized in the following theorem:

Theorem 6.3.4.1. *Error results for matrix-vector multiplication. Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$ and consider the assignment $y := Ax$ implemented via dot products as expressed in [\(6.3.4\)](#). Then these equalities hold:*

$$\textbf{R-1B} \quad \tilde{y} = (A + \Delta A)x, \text{ where } |\Delta A| \leq \gamma_n |A|.$$

$$\textbf{R-1F} \quad \tilde{y} = Ax + \tilde{\delta}y, \text{ where } |\tilde{\delta}y| \leq \gamma_n |A| |x|.$$

Ponder This 6.3.4.1. In the above theorem, could one instead prove the result

$$\tilde{y} = A(x + \hat{\delta}x),$$

where $\hat{\delta}x$ is "small"?

▼ [Solution](#)

The answer is "sort of". The reason is that for each individual element of y

$$\tilde{\psi}_i = \tilde{a}_i^T (x + \hat{\delta}x)$$

which would appear to support that

$$\begin{pmatrix} \tilde{\psi}_0 \\ \tilde{\psi}_1 \\ \vdots \\ \tilde{\psi}_{m-1} \end{pmatrix} = \begin{pmatrix} \tilde{a}_0^T (x + \hat{\delta}x) \\ \tilde{a}_1^T (x + \hat{\delta}x) \\ \vdots \\ \tilde{a}_{m-1}^T (x + \hat{\delta}x) \end{pmatrix}.$$

However, the δx for each entry $\tilde{\psi}_i$ is different, meaning that we cannot factor out $x + \delta x$ to find that $\tilde{y} = A(x + \delta x)$.

However, one could argue that we know that $\tilde{y} = Ax + \delta y$ where $|\delta y| \leq \gamma_n |A| |x|$. Hence if $A\delta x = \delta y$ then $A(x + \delta x) = \tilde{y}$. This would mean that δy is in the column space of A . (For example, if A is nonsingular). However, that is not quite what we are going for here.